GROUP 7 | SMART CAMPUS

Energy Saving Subsystem Handover

Technical handover for the Crowd Management teammate

Purpose
Explain the working energy-saving prototype, reserved pins, sensor thresholds, HVAC test scenarios, MQTT topics and the exact integration points needed to replace the temporary crowd-level input with the Crowd Management output.

CONTROLLER • Arduino Uno WiFi Rev2

SENSORS • PIR / DHT11 / LDR

OUTPUT • LED lighting + simulated HVAC status

MQTT • HiveMQ public broker for prototype testing

HANDOVER STATUS

Core prototype working: PIR-based occupancy detection, LDR-based daylight detection, vacancy timer, LED control, DHT11 readings, provisional HVAC decision logic, phone-controlled simulation and MQTT publishing/subscribing have been tested. Threshold validation, broader fault diagnostics, Google Sheets logging and final group integration remain to be completed.

Code files being passed separately (the .ino files themselves are not embedded in this PDF):

- All_sensors_test_smart_energy_system - all physical sensors together; no HVAC logic.
- Smart_energy_system_no_mqtt - combined lighting and HVAC logic; runs without MQTT.
- Smart_energy_system_mqtt - combined lighting and HVAC logic with MQTT controls and telemetry.

Use the MQTT version for the current online integrated prototype. Use the no-MQTT version for offline circuit testing. Use the all-sensors test version for hardware troubleshooting.

1. What is working

The energy subsystem has two separate decision paths. They share the Arduino but should remain logically independent: lighting is based on PIR occupancy and LDR brightness; HVAC is based on the DHT11 environmental reading and a discrete crowd level supplied manually for testing.

Feature	Inputs	Output / behaviour	Current status
Smart lighting	PIR + LDR	LED ON only when occupied and dark. Turns OFF after vacancy timeout.	Working
Environmental monitoring	DHT11	Temperature (°C) and relative humidity (%) read approximately every 2 seconds.	Working
HVAC decision logic	Temperature + humidity + crowd level 0-3	Calculates OFF / LOW / LOW-MEDIUM / MEDIUM / MEDIUM-HIGH / HIGH.	Implemented; thresholds provisional
Phone simulation	MQTT mode / scenario / crowd commands	Phone can select LIVE/SIMULATION, environment scenario and crowd level.	Working
MQTT telemetry	Energy system state	Publishes environmental, lighting, occupancy and HVAC values.	Working in prototype testing

2. Hardware and pin allocation

The following pins are already used on the Energy Saving Arduino Uno WiFi Rev2. If Crowd Management is to share this same physical Arduino, these pins must not be reassigned without coordinating a combined pin map.

Arduino pin / rail	Energy component	Connection / purpose
D2	PIR output	Digital motion signal from the HW-416 PIR module.
D3	DHT11 data	Data pin from the 3-pin DHT11 module.
D5	LED	LED anode through a current-limiting resistor; LED cathode to GND.
A0	LDR analogue input	Junction between one LDR leg and the fixed resistor.
5V	PIR + DHT11 + LDR divider	Common 5V supply rail.
GND	All components	Common ground rail.

Pin conflict rule: If Crowd Management runs on the same Uno, do not use D2, D3, D5 or A0. Also coordinate D0/D1 (Serial) and any I2C pins/modules before allocating other pins. If Crowd Management uses a separate Arduino, its pin numbers are local to that board and do not electrically conflict with this Energy board.

Wiring details

- PIR HW-416: VCC to 5V, OUT to D2, GND to GND.
- DHT11 3-pin module: VCC to 5V, DATA to D3, GND to GND. The module form shown has onboard supporting components.
- LED: D5 through a series current-limiting resistor to the LED long leg (+); short leg (-) to GND. Confirm resistor value before rebuilding the circuit.
- LDR divider: one LDR leg to 5V; the other LDR leg joins A0 and one end of the fixed resistor; the other resistor end goes to GND.
- All components share a common GND. Disconnect USB power before moving wires.

3. Smart lighting: logic and test results

Lighting is controlled independently from HVAC. The PIR detects movement, while the LDR measures ambient brightness. The LED represents the room lighting.

Decision rule

LED ON = room is occupied AND ambient light is dark.
If the room is vacant, the LED remains OFF regardless of brightness. If the room is bright enough, the LED remains OFF even when occupancy is detected.

PIR occupancy	LDR ambient state	LED output
Vacant	Dark	OFF
Vacant	Bright	OFF
Occupied	Dark	ON
Occupied	Bright	OFF

Vacancy timer

- PIR hardware output hold time was adjusted to approximately 10 seconds during testing.
- Arduino software vacancy timeout is set to 3 seconds for rapid demonstration/testing.
- The software timer starts after the PIR output returns LOW. Therefore the LED may switch OFF about 13 seconds after the last movement (approximately 10 seconds hardware hold + 3 seconds software timeout).
- The software uses millis() rather than a long delay for the vacancy countdown, so the main loop can continue reading other sensors.

LDR calibration observations

Lighting condition	Observed raw LDR reading
Fully covered / dark, LED OFF	0
Fully covered / dark, LED ON	4
Normal room lighting, LED OFF	20
Normal room lighting, LED ON	24

Current LDR thresholds: DARK at 8 or below; BRIGHT at 16 or above. Between 9 and 15, the previous state is retained (hysteresis) to reduce rapid switching. The LDR was initially affected by the prototype LED shining onto it. Repositioning/shielding the LDR from the LED resolved the flicker. Keep the LDR exposed to room light while preventing direct illumination from the prototype LED.

PIR test observations

Test	Observed result
Walk across sensor at 1 m	Passed
Walk across sensor at 2 m	Not reliable; accepted 1 m as the prototype working range
Stop moving after detection	Output returned LOW after approximately 10 seconds
Repeat movement after reset	Passed

Interpretation: The PIR is a motion sensor, not a people counter or a guaranteed continuous-presence detector. A still person may not generate fresh motion events. The vacancy timer reduces immediate false vacancy, but the 1 m range and timer are prototype settings, not building-deployment specifications.

4. DHT11 and HVAC decision logic

The DHT11 supplies actual temperature and relative humidity. For HVAC tests, the phone can switch between LIVE mode (actual DHT11 data) and SIMULATION mode (fixed test values). A crowd level is represented as an integer from 0 to 3.

Actual DHT11 observations

Observation	Temperature	Humidity
Normal room reading (earlier sample)	30.40 °C	73.30% RH
Later live reading example	30.10 °C	75.50% RH

Current provisional threshold definitions

Measurement	Current rule	Meaning
Temperature	High when >= 31.0 °C	Otherwise classified as not high
Relative humidity	High when >= 75.0% RH	Otherwise classified as not high
Environment category	HIGH if temperature OR humidity is high	NORMAL only if both are below their high thresholds

Threshold caveat: 31 °C and 75% RH are provisional prototype thresholds selected around observed room readings. They have not been fully validated as comfort, HVAC or energy-optimal thresholds. A live reading of 30.10 °C and 75.50% RH is classified HIGH because humidity is at or above 75%. In simulation, NORMAL is a selected scenario input (25 °C / 55% RH), regardless of the physical DHT11 reading.

Phone simulation presets

Phone scenario payload	Simulated temperature	Simulated humidity	Expected environment
NORMAL	25 °C	55% RH	NORMAL
HOT	33 °C	55% RH	HIGH
HUMID	25 °C	85% RH	HIGH
HOT_HUMID	33 °C	85% RH	HIGH

Crowd level mapping

Payload	Meaning
0	NONE
1	LOW
2	MEDIUM
3	HIGH

5. HVAC decision table and test scenarios

The decision table follows the group IoT Project 2 planning document. The current code implements its detailed HVAC labels and also publishes a simplified four-level label for dashboards that only support OFF / LOW / MEDIUM / HIGH.

Crowd level	Environment	Detailed HVAC setting	Standard output
NONE (0)	NORMAL	OFF	OFF
NONE (0)	HIGH	OFF / standby	OFF
LOW (1)	NORMAL	LOW	LOW
MEDIUM (2)	NORMAL	LOW-MEDIUM	MEDIUM
HIGH (3)	NORMAL	MEDIUM	MEDIUM
LOW (1)	HIGH	MEDIUM	MEDIUM
MEDIUM (2)	HIGH	HIGH	HIGH
HIGH (3)	HIGH	HIGH	HIGH
HIGH (3) + high temp / normal RH	Special case	MEDIUM-HIGH	HIGH

The special MEDIUM-HIGH case is triggered when crowd is HIGH, temperature is high and humidity is not high. The standard output is mapped to HIGH so the four-level MQTT output remains compatible. Always use the detailed topic when checking the exact table result.

Recommended validation matrix

Test	Scenario	Crowd	Expected detailed result
1	NORMAL	0 - NONE	OFF
2	HOT	0 - NONE	OFF
3	NORMAL	1 - LOW	LOW
4	NORMAL	2 - MEDIUM	LOW-MEDIUM
5	NORMAL	3 - HIGH	MEDIUM
6	HOT	1 - LOW	MEDIUM
7	HOT	2 - MEDIUM	HIGH
8	HOT	3 - HIGH	MEDIUM-HIGH
9	HUMID	1 - LOW	MEDIUM
10	HOT_HUMID	3 - HIGH	HIGH

Important current behaviour: PIR occupancy controls the lighting path; the HVAC path uses the discrete crowdLevel value. Therefore PIR may show OCCUPIED while HVAC remains OFF if crowdLevel is 0. This is the current code behaviour, not an error. The group should decide whether the integrated design should gate HVAC with PIR occupancy, crowd data, or both.

6. MQTT configuration and topics

The working MQTT sketch uses the HiveMQ public broker for prototype testing. The phone can send operating mode, scenario and crowd commands; the Energy Arduino subscribes to these commands and publishes its readings and decisions.

Setting	Value
Broker	broker.hivemq.com
Port / protocol	1883 / TCP
Username / password	None
TLS	Off
Energy Arduino Client ID	sitg7-energy-uno02
Android panel Client ID	sitg7-phone-01

Incoming command topics (phone to Energy Arduino)

Topic	Payloads
sitg7/room1/cmd/mode	LIVE / SIMULATION
sitg7/room1/cmd/scenario	NORMAL / HOT / HUMID / HOT_HUMID
sitg7/room1/cmd/crowd	0 / 1 / 2 / 3 (phone simulation/testing)

Current energy telemetry topics (Energy Arduino to broker)

Topic	Payload
sitg7/room1/energy/temp	Effective temperature used by HVAC
sitg7/room1/energy/hum	Effective humidity used by HVAC
sitg7/room1/energy/actual_temp	Actual DHT11 temperature
sitg7/room1/energy/actual_hum	Actual DHT11 humidity
sitg7/room1/energy/crowd	Crowd level currently used (0-3)
sitg7/room1/energy/hvac	Standard OFF / LOW / MEDIUM / HIGH output
sitg7/room1/energy/hvac_detail	Detailed HVAC label, including intermediate levels
sitg7/room1/energy/lighting	ON / OFF
sitg7/room1/energy/occupancy	OCCUPIED / VACANT
sitg7/room1/energy/ambient	DARK / BRIGHT
sitg7/room1/energy/mode	LIVE / SIMULATION
sitg7/room1/energy/scenario	Selected scenario payload
sitg7/room1/energy/dht_status	NORMAL / FAULT

The Android Text Output diagnostic panel can subscribe to sitg7/room1/# to see all subtopics. The three Combo Box controls publish to their respective cmd topics; dashboard prefix must be disabled so the exact topic is used.

7. Crowd Management integration: what to edit

The most important handover item is the boundary between the Crowd Management output and the Energy HVAC input. Crowd Management must convert its own people-count / occupancy estimate into one discrete level before that value is used by the Energy decision logic.

A. If both subsystems run on the same Arduino

- Keep one shared variable named crowdLevel (or map the Crowd code output into this existing variable). Do not create a second crowdLevel variable with a different value.
- Map the crowd algorithm output to 0, 1, 2 or 3 using crowd-count bands agreed by the group. The actual numeric boundaries are not defined in the Energy code and must be decided by Crowd Management.
- Update crowdLevel before updateHVAC() / calculateHVAC() runs. The HVAC function already consumes crowdLevel.
- Preserve the Energy pin assignments: D2, D3, D5 and A0. Create one combined pin allocation table with the hardware integration owner before adding crowd sensors or an LCD.

Example mapping pattern (replace placeholder count limits with the Crowd team's agreed thresholds):

```
// Example only - agree the count boundaries first.
int crowdLevel;

if (estimatedPeople == 0) {
  crowdLevel = 0;        // NONE
} else if (estimatedPeople <= LOW_MAX) {
  crowdLevel = 1;        // LOW
} else if (estimatedPeople <= MEDIUM_MAX) {
  crowdLevel = 2;        // MEDIUM
} else {
  crowdLevel = 3;        // HIGH
}

// The Energy HVAC logic reads this same crowdLevel.
```

B. If Crowd Management runs on a separate Arduino

- The Crowd Arduino should publish a numeric payload 0, 1, 2 or 3 using the group topic already proposed by the Crowd teammate: sitg7/room1/crowd. Publish as a simple integer payload, not a JSON object, unless both teams agree to change the parser.
- The Energy Arduino must subscribe to sitg7/room1/crowd in addition to its existing command subscription sitg7/room1/cmd/#.
- In the Energy MQTT callback, add a handler for the exact incoming topic sitg7/room1/crowd; validate that the payload is 0-3, then assign it to the existing crowdLevel variable.
- The phone command topic sitg7/room1/cmd/crowd is for simulation/testing. Recommended source selection: accept the phone crowd command only when mode is SIMULATION; accept the Crowd board topic only when mode is LIVE. This prevents two sources from overwriting crowdLevel unpredictably.

Code integration pattern for the Energy MQTT subscriber:

```
// In subscribeCommands(), add:
mqttClient.subscribe("sitg7/room1/crowd");

// In onMqttMessage(), use the right source by mode:
else if (simulationMode && topic.endsWith("/cmd/crowd")) {
  applyCrowdLevel(value);  // phone test selector
}
else if (!simulationMode && topic == "sitg7/room1/crowd") {
  applyCrowdLevel(value);  // live Crowd board
}

// Helper: accept only the agreed 0-3 payloads
void applyCrowdLevel(String value) {
  if (value == "0" || value == "1" ||
      value == "2" || value == "3") {
    crowdLevel = value.toInt();
    Serial.print("Crowd level updated: ");
    Serial.println(crowdLevel);
  }
}
```

8. Restart, test and handover checklist

Restart procedure

- Turn on the phone hotspot (2.4 GHz compatible) and ensure it has internet access.
- Power the Arduino and upload/run the MQTT sketch. Wait for WiFi connected, MQTT connected and command subscription confirmation in Serial Monitor.
- Open IoT MQTT Panel connection SIT Group 7 on Android. Its saved connection has Auto connect enabled; if the cloud icon shows disconnected, tap it to reconnect.
- After Arduino restart, reselect LIVE or SIMULATION, then scenario and crowd level. Defaults reset to LIVE, NORMAL and crowd 0; commands are not retained.
- Check Text Output subscriptions or the diagnostic topic sitg7/room1/# to confirm fresh messages are arriving.

Suggested integration test order

Order	Test	Pass condition
1	Run All_sensors_test_smart_energy_system	PIR, LDR, DHT11 and LED report readings / behave locally.
2	Run Smart_energy_system_no_mqtt	Lighting and HVAC decisions work without Wi-Fi/MQTT.
3	Run Smart_energy_system_mqtt	MQTT connection, subscriptions and telemetry work.
4	Crowd mapping	Crowd output maps consistently to 0/1/2/3.
5	Cross-board or same-board integration	Energy HVAC consumes the Crowd Management level, not a duplicate/manual value in LIVE mode.
6	Full demo	Run NORMAL, HOT, HUMID and HOT_HUMID scenarios with all crowd levels; verify detailed HVAC labels.

Known limitations / items still to complete

- HVAC thresholds (31.0 °C and 75.0% RH) are provisional and require further validation / justification.
- PIR range was reliable at 1 m in the test; 2 m test was not reliable. PIR is motion detection, not people counting.
- PIR and LDR comprehensive fault detection is not implemented. DHT11 invalid/NaN readings are handled; detailed fault handling remains limited.
- The physical LED is a lighting representation only. HVAC is simulated as text/status output; the prototype does not control actual air-conditioning equipment.
- Google Sheets logging, historical energy-saving event logging and final dashboard visualisation are not yet completed.
- A manual override button was not available in the current hardware set.

Code file handover: recommended use

File supplied separately	Use it for
All_sensors_test_smart_energy_system	Hardware-only diagnosis of PIR, LDR, DHT11 and LED; no HVAC logic.
Smart_energy_system_no_mqtt	Offline / standalone test of lighting plus HVAC decision logic.
Smart_energy_system_mqtt	Current MQTT-enabled integrated Energy prototype; use as the starting point for group MQTT integration.

Handover request: When editing the Crowd sketch, please share (1) the chosen crowd-count thresholds for NONE/LOW/MEDIUM/HIGH, (2) whether the Crowd and Energy code will run on the same Arduino or separate boards, (3) the final pin allocation, and (4) the agreed MQTT topic for the live crowd value.